Robot Learning for Complex Manufacturing Process

Heping Chen, Binbin Li, Dave Gravel, George Zhang and Biao Zhang

***Abstract*—At the present time, industrial robots for assembly tasks only constitute a small portion of the annual robot sales. One of the main reasons is that it is difficult for conventional industrial robots to adapt to the complicity and flexibility of assembly processes. Therefore, intelligent industrial robotic systems are attracting more and more attention. However, because of the modeling difficulty and low efficiency of the existing solutions, optimal performance is difficult to achieve. In this paper, a parameter learning method is developed based on Gaussian Process Regression Bayesian Optimization Algorithm (GPRBOA). Gaussian Process Regression(GPR) is utilized to model the relationship between the process parameters and system performance. GPRBOA is proposed to optimize the process parameters. The experiments were performed using a complex three stage torque converter assembly process. The experimental results verify the effectiveness of the robot learning method and demonstrate its efficiency compared to Design Of Experiment(DOE) methods. The proposed method can greatly increase the manufacturing efficiency and will generate big economic impact.**

I. Introduction

In the manufacturing processes, the environment is constantly changing and parts to be processed could come from different batches and sometimes different suppliers. All of these variations will cause difficulty for conventional industrial robots to perform many manufacturing processes[1], [2], for example, the torque converter assembly in the power train assembly process [3], [4] as shown in Figure 1.

Automotive torque converter assembly is a typical robotic assembly process. Torque converter can weigh up to 75 kilograms. Between torque converter and transmission house, there typically exist several distinct subassembly stages to be assembled during the assembly process. Assembling torque converter manually causes working injury and other ergonomic problems. Automating torque converter assembly is demanded by the automotive industry. Robotic force control provides a better solution than hard automation equipment with more flexibility and cost advantages. From Figure 1, we can see that the torque converter assembly is an insertion process with several layers of gear matching. Experience tells that adding hopping motion along the axial direction will help improving the assembly cycle time because the constant axial insertion force will press the two assembling surfaces together, which prevents gear matching during assembly. Installing a converter into a transmission is not always as easy as it looks. Once the converter begins to engage the transmission, it should be centered with the hole in the transmission. At the same time, the converter has to be rotated back and forth and hopped up and down repeatedly to allow the splines and hub tabs to engage the transmission correctly. Because there are several such distinct stages in the converter to be assembled into the transmission before it is all the way in, many parameters are involved in this assembly process, such as rotation speed and angle, hopping speed and amplitude and insertion force for each stage. The assembly process performance will decrease if these parameters are not tuned correctly to adapt to the variations.

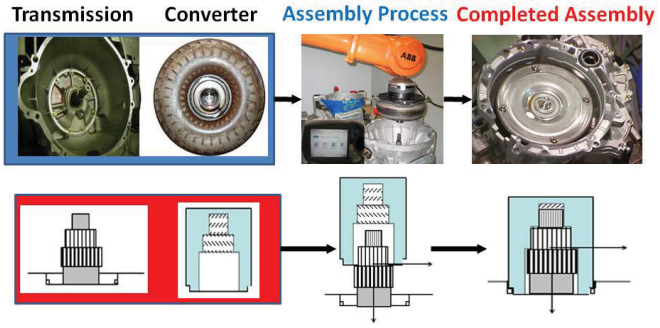


Fig. 1. The torque converter assembly process.

Several offline algorithms have been proposed to solve the assembly process parameter optimization problem[5], [6]. The Genetic algorithms (GA) are developed to randomly search for optimal parameters[7], [8]. To increase the efficiency of the GA based methods, Artificial Neural Network (ANN) is utilized to model whether the parameters are "good" or "bad" to filter the candidate parameters first without performing any experiment[9]. The design-of-experiment (DOE) methods[10], [11] adopt a systematic way to optimize parameters. After performing a series of experiments, the most sensitive parameters are chosen and tuned carefully[12], [3]. Even though these methods are effective in offline parameter optimization, it is unreasonable to use them online because of their low efficiency.

To our best knowledge, no practical online parameter optimization algorithm has been proposed for complex torque converter assembly process. Because the assembly processes typically have many stages and different control strategies[13] such as hopping and searching[14], [15], it is hardly possible to construct a physical model using first principles[16] to optimize the process parameters. Therefore an online parameter optimization method without using any physical model has to be investigated.

In this paper, a robot learning method based on Gaussian Process Regression surrogated Bayesian Optimization Algorithm (GPRBOA) is proposed to solve the parameter optimization problem online. Experiments performed using the complex torque converter process demonstrate the effectiveness of the proposed algorithm. Compared to the DOE based methods, the efficiency and accuracy are greatly improved; moreover it can be applied online without stopping an assembly station to perform experiments for parameter optimization.

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II. Proposed Solution

A high precision robotic assembly requires a robot to perform assemblies in which the assembly clearance is close to or better than the robots' repeatability. The clearance and geometry of parts from different batches, sometime different suppliers, are different. These variations will cause the increase of cycle time; moreover, assembly failure if the assembly process parameters do not change. To deal with the problem, the assembly process parameters should be tuned to adapt to the variations. However, it is difficult to tune all these parameters since the relationship between the parameters and system performance is not clear. Thus, we propose a parameter learning method to optimize the assembly process parameters based on system performance evaluation. Figure 2 shows the framework for assembly process parameter optimization.

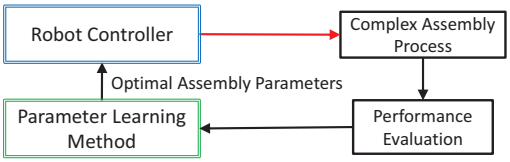


Fig. 2. Framework of automated assembly parameter optimization.

For a complex assembly system, the performance of the system can be evaluated after an assembly is performed. The performance could include several indices such as cycle time and success rate etc. Together with the current assembly parameters, it is input to the parameter optimization algorithm. The algorithm will generate a new set of assembly parameters to control the complex assembly system. The process continues until optimal system performance is achieved with a set of optimal assembly parameters. Figure 3 shows the procedures of the optimization process.

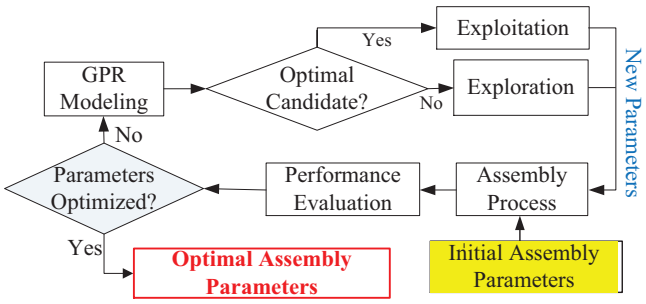


Fig. 3. Assembly parameter optimization process based on Gaussian process regression Bayesian optimization algorithm.

Using the initial set of parameters, experiment is performed and the system response is recorded. The system performance is then evaluated. If the system performance satisfies given requirements, optimal assembly parameters are found; otherwise, the data are used to update the model using Gaussian Process Regression. A balancing function based on Bayesian optimization is then used to determine either the exploration process or exploitation process will be deployed to find the new assembly parameter set. Experiment is then performed again using the new assembly parameters. The process continues until the assembly parameters are optimized.

A. Gaussian Process Regression (GPR)

Gaussian Process Regression is a non-parametric tool that can handle the modeling problem with noisy observations and system uncertainties[17]. It is widely used to model geographical terrains[18], dynamical systems[19], nonlinear systems[20], complex environments[21] and sensor networks[22] etc.

For a Gaussian Process $f(\mathbf{x})$, a set of multivariate Gaussian random variables $\mathscr{F} = \{f(\mathbf{x}_1), f(\mathbf{x}_2), , ..., f(\mathbf{x_N})\}$ can be defined over $\mathscr{X}$ with any finite set of N points $\{\mathbf{x}_i \in \mathscr{X}\}_{i=1}^{N}$ where $f(\mathbf{x_i})$ has the value of the latent function $f(\mathbf{x})$ at $\mathbf{x_i} \in \mathscr{X}$ and $\mathscr{X}$ is defined over $\mathbb{R}^D$. $f(\mathbf{x})$ is completely specified by its mean function $m(\mathbf{x})$ and covariance function $k(\mathbf{x},\mathbf{x}')$: $f(\mathbf{x}) \sim \mathscr{GP}(m(\mathbf{x}),k(\mathbf{x},\mathbf{x}'))$ where $\mathbf{x}$ and $\mathbf{x}'$ are two arbitrary variables in $\mathscr{X}$.

For a model $y = f(\mathbf{x}) + \mathbf{w}$ and $w \sim N\left(0, \sigma_n^2\right)$, the covariance function is $\mathrm{cov}\left(y_i, y_j\right) = k\left(\mathbf{x}_i, \mathbf{x}_j\right) + \sigma_n^2 \delta_{ij}$, where the δ_{ij} is the Kronecker delta which is one iff $i = j$ and zero otherwise. The joint distribution of the observed data set $(\mathbf{X}, \mathbf{y})$ and predicted data set $(\mathbf{X}_*, \mathbf{y}_*)$ is

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{y}_* \end{bmatrix} \sim N\left(0, \begin{bmatrix} K(\mathbf{X},\mathbf{X}) + \sigma_n^2 I & K(\mathbf{X},\mathbf{X}_*) \\ K(\mathbf{X}_*,\mathbf{X}) & K(\mathbf{X}_*,\mathbf{X}_*) \end{bmatrix}\right) \tag{1}$$

where $K(\mathbf{X}', \mathbf{X}'')$ ($\mathbf{X}'$ and $\mathbf{X}''$ refer to $\mathbf{X}$ and $\mathbf{X}_*$) is the covariance matrix whose element K_{ij} in i_{th} row and j_{th} column equals to $k(\mathbf{x}'_\mathbf{i}, \mathbf{x}''_\mathbf{j})$. By deriving the conditional distribution, we can obtain the predictive function as

$$\mathbf{y}_* | \mathbf{X}, \mathbf{y}, \mathbf{X}_* \sim N\left(\mu(\mathbf{y}_*), \mathrm{V}(\mathbf{y}_*)\right) \tag{2}$$

$$\begin{aligned} \mu(\mathbf{y}_*) &= E\left[\mathbf{y}_* | \mathbf{X}, \mathbf{y}, \mathbf{X}_*\right] = K(\mathbf{X}_*, \mathbf{X})\left[K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 I\right]^{-1} \mathbf{y} \\ \mathrm{V}(\mathbf{y}_*) &= K(\mathbf{X}_*, \mathbf{X}_*) - K(\mathbf{X}_*, \mathbf{X})\left[K(\mathbf{X}, \mathbf{X}) + \sigma_n^2 I\right]^{-1} K(\mathbf{X}, \mathbf{X}_*) \end{aligned} \tag{3}$$

where $\mu(\mathbf{y}_*)$ is the predicted mean for $\mathbf{y}_*$ and $\mathrm{V}(\mathbf{y}_*)$ is the predicted variance. The covariance function $K(\mathbf{x},\mathbf{x}')$ is a key for determining a GPR model. Among various forms of covariance functions, the commonly-used one is the Squared Exponential $k_{\mathrm{SE}}(\mathbf{x},\mathbf{x}')$ which is

$$k_{\mathrm{SE}}(\mathbf{x},\mathbf{x}') = \sigma^2 \exp\left(-\frac{|\mathbf{x}-\mathbf{x}'|^2}{2l^2}\right) \tag{4}$$

where l is a characteristic length-scale factor controlling how close $\mathbf{x}$ and $\mathbf{x}'$ are and σ^2 is the amplitude of the variance. For each covariance function, there are some hyperparameters θ such as l and σ for k_{SE}. The goal of model construction is to find a covariance function $k(\mathbf{x},\mathbf{x}',\theta)$ which fits the data set (X,y) best. Suppose $f(\mathbf{x})$ is a candidate latent function(f in short) for the given data set, the posterior probability is

$$p(f|X,y,H,\theta) = \frac{p(y|X,f,H,\theta)\, p(f|X,H,\theta)}{p(y|X,H,\theta)} \tag{5}$$

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where H is the hypothesis on the structure of the covariance function, θ is the hyperparameters, X, y are the sample data sets and

$$p(y|X,H,\theta) = \int p(y|X,f,H,\theta)\, p(f|X,H,\theta)\, df \qquad (6)$$

is the marginal likelihood that refers to the marginalization over the function f.

As mentioned above, in the Gaussian Process Regression, the function $f(\mathbf{x})$ is not given explicitly. Thus, $p(y|X,H,\theta)$ actually refers to the likelihood of H and θ given the data set (X,y). The Gaussian assumption makes it possible to derive the analytical solution of the log marginal likelihood

$$\begin{aligned} \log p(y|X,H,\theta) &= -\tfrac{1}{2}\mathbf{y}^{\mathrm{T}}\left(K+\sigma_n^2 I\right)^{-1}\mathbf{y} \\ &- \tfrac{1}{2}\log\left|K+\sigma_n^2 I\right| - \tfrac{n}{2}\log 2\pi \end{aligned} \qquad (7)$$

Therefore the hyper-parameters θ in a covariance function H can be optimized by maximizing the marginal log likelihood:

$$\theta^* = \underset{\theta}{argmax} \log p(\mathbf{y}|\mathbf{X},H,\theta) \qquad (8)$$

This optimization problem can be solved using different techniques such as Conjuncted Gradient Algorithm or evolutionary algorithms. In this paper, Nelder-Mead(Simplex) method is utilized which is effective and uses only the value of $\log p(y|X,H,\theta)$. Once $k(\mathbf{x},\mathbf{x}',\theta^*)$ is known, equation (3) can be used to make prediction for new input $\mathbf{x}^*$.

B. Bayesian Optimization Algorithm

The Bayesian Optimization Algorithm(BOA) estimates a probability distribution of promising solutions using Bayesian Network(BN) in order to generate new candidate solutions. The BN model is updated at each iteration using new samples. The BOA achieves a good balance between the modeling difficulty and parameter optimization efficiency and has been applied to solve many optimization problems. However, in each iteration both the BN structure and parameters have to be optimized and this optimization process will increase the computational complexity and difficulty. Therefore, in this paper, the BN is replaced with the GPR model.

The GPR surrogated BOA (GPRBOA) can iteratively model a complex system and optimize the system performance. In each iteration, new samples are added into the existing data set and used to update the Gaussian process model. The new model will then be used to search for the optimal solutions by maximizing a performance index $s(x)$(also known as the acquisition function) over the input domain. Because the acquisition function controls the new sample points, it directly affects the quality of the built model and the optimal solution. Therefore, it plays a key role in the GPRBOA algorithm. Typically the acquisition function defined using $\mu(\mathbf{x})$ and $\sigma(\mathbf{x})$ is deployed to acquire a new candidate using different techniques such as Probability of Improvement, Expected Improvement and Lower Confidence Bound(LCB).

LCB is recently developed based on the idea of exploiting lower confidence bounds (upper, when considering maximization) to increase the optimization efficiency. It has the form

$$LCB(\mathbf{x}) \overset{\Delta}{=} \mu(\mathbf{x}) - \kappa \cdot \sigma(\mathbf{x}) \qquad (9)$$

where κ is a scaling factor. Instead of only sampling the points with minimum mean $\mu(\mathbf{x})$ or maximum variance $\sigma(\mathbf{x})$ predicted using the current model to improve the model uncertainty, LCB reduces the search space by combining the mean with variance, which can ignore points that has no possibility of being optimal.

C. GPRBOA-VR Algorithm

The difficulties of deploying GPRBOA online come from the following two aspects: the variations of the assembly process and computational burden of GPR[23].

1) Variations of the assembly process: GPRBOA performs online modeling and optimization simultaneously using the exploration and exploitation processes. If the exploration and exploitation processes are not properly balanced, the optimization process can be trapped in local minima. Hence the proposed algorithm must be able to balance the two processes. LCB method can explore a system by sampling $\mathbf{x}$ with large $\mu(\mathbf{x})$ and exploit the model by sampling $\mathbf{x}$ with large $\sigma(\mathbf{x})$. Hence it requires prior information about the variance of cycle time. However, for different batches or assembly processes, such prior information is not available. To deal with such a problem, we propose a new algorithm called GPRBOA-VR to balance the exploration and exploitation processes:

$$\mathbf{x}^* = \begin{cases} \arg\min LCB(\mathbf{x}) & rand(1) > \varepsilon \\ \arg\max VR(\mathbf{x}) & otherwise \end{cases} \qquad (10)$$

where ε is the performance index which is updated online according to the hyper-parameters and system performance; $VR(\mathbf{x})$(Variation Random) is a random acquisition function used to explore the unsampled area to improve the model quality by investigating the farthest unsampled points:

$$VR(\mathbf{x}) = \min_{i \in 1 \cdots N} [d(\mathbf{x} - \mathbf{x_i})] \qquad (11)$$

where $d(\mathbf{x}-\mathbf{x}')$ is the distance between two sets of parameters $\mathbf{x}$ and $\mathbf{x}'$. At each iteration, if rand$(1) > \varepsilon$, the new candidate is optimized by exploiting the current model; otherwise, it is calculated by exploring the unsampled parameter space. Hence the exploitation process optimizes the process parameters according to the constructed GPR model while the exploration process refines the model according to random variation.

III. Experimental Results

Experiments were performed to demonstrate the efficiency of the proposed method using a torque converter assembly process as shown in Figure 4. The experimental system consists of an ABB IRB4400 robot with an IRC5 controller and a force sensor mounted on the robot endeffector. A computer is used to run the parameter optimization algorithm and communicate with robot controller via ethernet. ABB force control package is used to perform the complex assembly process.

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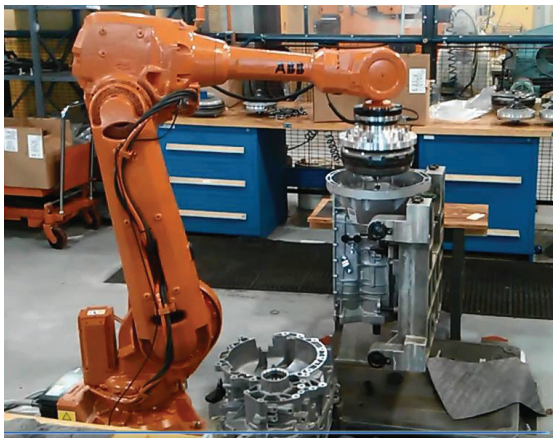


Fig. 4. The experimental system using an ABB IRB4400 for torque converter assembly.

Based on the nature of common assembly types, assembly processes are categorized into cylindrical, radial and multi-stage insertion/assembly processes. Cylindrical insertion includes assembly applications with a cylinder and a smooth bore, such as transmission torque converter assembly. Spiral search is used as primary pattern. Radial insertion includes assembly applications with a toothed, splined or gear mesh, such as a forward clutch assembly and spline gear assembly. Rotational search is used as primary pattern. And multi-stage insertion which requires the combination of cylindrical and radial at different stages includes assembly applications such as a torque converter assembly, half shaft-differential assembly and the like. The combination of rotational and circular motion is often used as search pattern. Besides, hopping, oscillating along the insertion direction, can help a lot to overcome the contact friction force and the sliding between the loosing parts. The cylindrical, radial and multi-stage insertion types cover most of the real-world assembly applications. For each insertion/assembly type, certain search pattern is applied to perform the assembly process. Correspondingly, there are certain robot force control parameters that are related to each search pattern. The following parameters are optimized in complex torque converter assembly process:

- *Insertion Distance:* distance between the insertion starting position and the end position
- *Engage Distance:* distance between the insertion starting position to the position where the parts are engaged
- *Search force:* Max value of the searching/insertion force used in the tool Z direction
- *Rotation Speed:* the rotation speed used in the insertion around the tool Z axis
- *Rotation Angle:* the maximum angle value used in the above rotation motion
- *Force Amplitude:* the amplitude of the hopping force in the Z direction of tool frame
- *Force Period:* the period of the hopping (oscillation) force

For this torque converter, there are three stages and each stage has seven parameters. A series of experiments were performed to optimize the parameters. Figure 5 shows a typical experimental result for the first stage. The optimal parameters were found after five experiments. Both exploration and exploitation processes were involved. After the optimal set of parameters was determined, production process started. In production, there are cycle time variations. These variations are caused by the part variations because we added some noises to the part position and orientation.

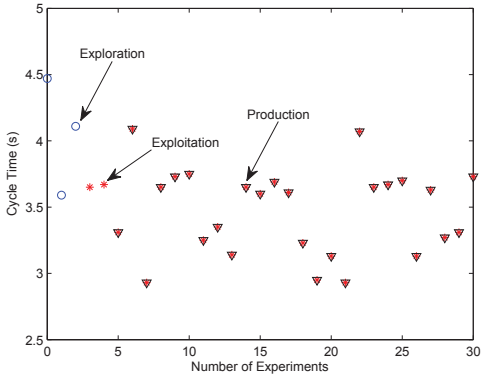


Fig. 5. The cycle time for the first stage. The first stage converges after 5 experiments

For the second and third stages, similar results were found. The second stage converges faster with only three experiments. There exists only exploration process. This is because the second stage is easier to assemble. The cycle time for the second stage is much less than the other two stages. The third stage took much longer time because the oil pump assembly is involved. During the assembly process, the orientation of the oil pump is intentionally changed. Because the oil pump only has two teeth, the robot must search for a bigger range in order to engage the oil pump.

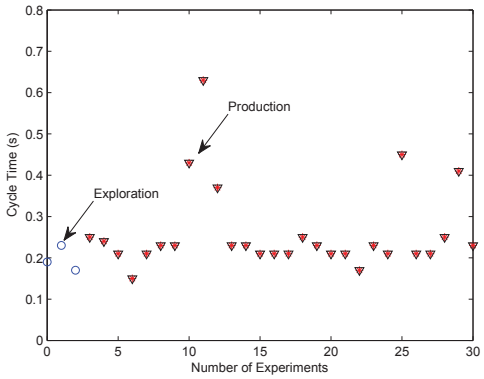


Fig. 6. The cycle time for the second stage. The second stage converges after three experiments

From the experimental results we can see that the proposed method can optimize the process parameters very efficiently for complex assembly process. Using DOE methods, many more experiments have to be performed in order to optimize 21 parameters in the three stage assembly process.

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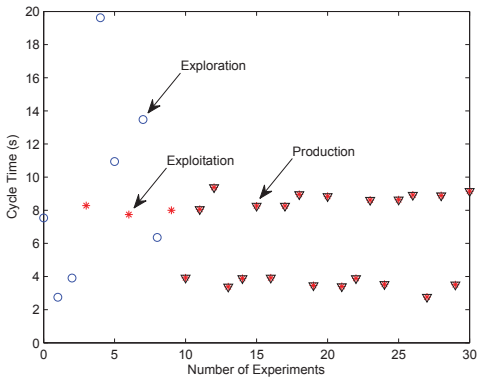


Fig. 7. The cycle time for the third stage. The third stage converges after 10 experiments

IV. Conclusion

The paper proposes a robot learning method based on Gaussian process regression Bayesian optimizetion algorithm to efficiently optimize the assembly parameters in a complex robotic assembly processes. The cycle time is optimized by balancing the exploration and exploitation processes. Compared to the existing methods, such as DOE and GA, GPRBOA can greatly improve the parameter optimization efficiency without stopping the production line for experiments. Experimental results demonstrate the effectiveness of the proposed method. The proposed robot learning method will generate big economic impact in automotive manufacturing. It can also be used to model complex processes, to optimize the process parameters and improve the system performance.

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